Learn Python by writing it. This book takes you from your first line of code to tested, packaged programs, one small, clear step at a time.

Every example is a real program, run on Python 3.12, 3.13, 3.14 and 3.15 before publication, with the exact output printed beneath it. Type what you see, and you'll get what you see.

You'll learn variables, decisions, loops, lists, dictionaries, functions, files and errors; then classes, testing with pytest, debugging, packaging, type hints, decorators, concurrency and performance. Then you'll put it all to work: a command-line app with a SQLite database, a data analysis with pandas and charts, a web API with FastAPI, and scripts that automate everyday tasks.

Every chapter ends with exercises, and every exercise has a tested, worked solution. No programming experience needed.

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PYTHON PROGRAMMING HANDS-ON

AADHAVAN MUTHURENGAN

shelf.py

```
1  def shelf(books):
2      for n, title in enumerate(books, 1):
3          print(f"{n}. {title}")
4
5  shelf(sorted(["Dune", "Emma"]))
```

```
$ python shelf.py
1. Dune
2. Emma
```

- BASICS
- FUNCTIONS
- CLASSES
- TESTING
- DATA
- WEB APIs

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FROM BASIC TO ADVANCED

Learn Python by Writing Real Programs, with Every Example Tested

120+ TESTED EXAMPLES • EXERCISES & SOLUTIONS • REAL PROJECTS

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